

1. Create Tables (oracle application express- SQL command)

2. Add constraints (PK, FK, etc)

```
CREATE TABLE Library_Member(  
  member_ID number(6) CONSTRAINT lib_mem_id_pk PRIMARY KEY,  
  first_name varchar2(40) NOT NULL,  
  last_name varchar2(40) NOT NULL,  
  phone varchar2(40),  
  email varchar2(40),  
  home_address varchar2(60) NOT NULL,  
  member_ship_detail varchar2(40),  
  CONSTRAINT lib_phone_email_uk UNIQUE (phone, email)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
MEMBER_ID	NUMBER(6,0)	No	-	1
FIRST_NAME	VARCHAR2(40)	No	-	-
LAST_NAME	VARCHAR2(40)	No	-	-
PHONE	VARCHAR2(40)	Yes	-	-
EMAIL	VARCHAR2(40)	Yes	-	-
HOME_ADDRESS	VARCHAR2(60)	No	-	-
MEMBER_SHIP_DETAIL	VARCHAR2(40)	Yes	-	-

```
CREATE TABLE Library_Book(  
  book_ID number(6) CONSTRAINT lib_book_id_pk PRIMARY KEY,  
  title varchar2(40) NOT NULL,  
  genre varchar2(40),  
  availability varchar2(40),  
  price varchar2(40) NOT NULL  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BOOK_ID	NUMBER(6,0)	No	-	1
TITLE	VARCHAR2(40)	No	-	-
GENRE	VARCHAR2(40)	Yes	-	-
AVAILABILITY	VARCHAR2(40)	Yes	-	-
PRICE	VARCHAR2(40)	No	-	-

```
CREATE TABLE Library_Preference(  
  preference_ID number(6) CONSTRAINT lib_pref_id_pk PRIMARY KEY,  
  preference_type varchar2(40)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
PREFERENCE_ID	NUMBER(6,0)	No	-	1
PREFERENCE_TYPE	VARCHAR2(40)	Yes	-	-

```
CREATE TABLE Library_Librarian(  
  librarian_ID number(6) CONSTRAINT lib_lib_id_pk PRIMARY KEY,  
  name varchar2(40),  
  member_ID number(6),  
  CONSTRAINT lib_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
LIBRARIAN_ID	NUMBER(6,0)	No	-	1
NAME	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-

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```
CREATE TABLE Library_Book_Order(  
    book_order_ID number(6) CONSTRAINT lib_book_ord_id_pk PRIMARY KEY,  
    order_date varchar2(40),  
    guarantee_payment varchar2(40),  
    status varchar2(40),  
    member_ID number(6),  
    book_ID number(6),  
    CONSTRAINT lib1_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),  
    CONSTRAINT lib1_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BOOK_ORDER_ID	NUMBER(6,0)	No	-	1
ORDER_DATE	VARCHAR2(40)	Yes	-	-
GARANTEE_PAYMENT	VARCHAR2(40)	Yes	-	-
STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Purchase(  
    purchase_ID number(6) CONSTRAINT lib_pur_id_pk PRIMARY KEY,  
    purchase_date varchar2(40),  
    payment_amount varchar2(40) NOT NULL,  
    member_ID number(6),  
    book_ID number(6),  
    CONSTRAINT lib2_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),  
    CONSTRAINT lib2_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
PURCHASE_ID	NUMBER(6,0)	No	-	1
PURCHASE_DATE	VARCHAR2(40)	Yes	-	-
PAYMENT_AMOUNT	VARCHAR2(40)	No	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Borrowing(  
    borrowing_ID number(6) CONSTRAINT lib_borr_id_pk PRIMARY KEY,  
    due_date varchar2(40),  
    return_date varchar2(40),  
    overdue_status varchar2(40),  
    member_ID number(6),  
    book_ID number(6),  
    CONSTRAINT lib3_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),  
    CONSTRAINT lib3_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
BORROWING_ID	NUMBER(6,0)	No	-	1
DUE_DATE	VARCHAR2(40)	Yes	-	-
RETURN_DATE	VARCHAR2(40)	Yes	-	-
OVERDUE_STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

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```
CREATE TABLE Library_Reservation(  
    reservation_ID number(6) CONSTRAINT lib_res_id_pk PRIMARY KEY,  
    request_date varchar2(40),  
    status varchar2(40),  
    member_ID number(6),  
    book_ID number(6),  
    CONSTRAINT lib4_mem_id_fk FOREIGN KEY (member_ID) REFERENCES Library_Member(member_ID),  
    CONSTRAINT lib4_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
RESERVATION_ID	NUMBER(6,0)	No	-	1
REQUEST_DATE	VARCHAR2(40)	Yes	-	-
STATUS	VARCHAR2(40)	Yes	-	-
MEMBER_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

```
CREATE TABLE Library_Author(  
    author_ID number(6) CONSTRAINT lib_aut_id_pk PRIMARY KEY,  
    name varchar2(40)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
AUTHOR_ID	NUMBER(6,0)	No	-	1
NAME	VARCHAR2(40)	Yes	-	-

```
CREATE TABLE Library_Book_Author(  
    author_ID number(6),  
    book_ID number(6),  
    CONSTRAINT lib_aut_id_fk FOREIGN KEY (author_ID) REFERENCES Library_Author(author_ID),  
    CONSTRAINT lib5_book_id_fk FOREIGN KEY (book_ID) REFERENCES Library_Book(book_ID)  
);
```

Column Name	Data Type	Nullable	Default	Primary Key
AUTHOR_ID	NUMBER(6,0)	Yes	-	-
BOOK_ID	NUMBER(6,0)	Yes	-	-

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3. Add data to tables (oracle application express)-more than five lines in each table

insert all into Library_Member

```
values (101, 'Emma', 'Stone', '1234567890', 'emma.stone@example.com', '123 Oak Street', 'Gold Member')
INTO Library_Member VALUES (102, 'Liam', 'Johnson', '2345678901', 'liam.johnson@example.com', '456 Pine Avenue',
'Silver Member')
INTO Library_Member VALUES (103, 'Olivia', 'Williams', '3456789012', 'olivia.williams@example.com', '789 Maple Blvd',
'Platinum Member')
INTO Library_Member VALUES (104, 'Noah', 'Brown', '4567890123', 'noah.brown@example.com', '321 Cedar Lane',
'Silver Member')
INTO Library_Member VALUES (105, 'Ava', 'Davis', '5678901234', 'ava.davis@example.com', '654 Birch Way', 'Gold
Member')
```

select 1 from dual;

MEMBER_ID	FIRST_NAME	LAST_NAME	PHONE	EMAIL	HOME_ADDRESS	MEMBER_SHIP_DETAIL
101	Emma	Stone	1234567890	emma.stone@example.com	123 Oak Street	Gold Member
102	Liam	Johnson	2345678901	liam.johnson@example.com	456 Pine Avenue	Silver Member
103	Olivia	Williams	3456789012	olivia.williams@example.com	789 Maple Blvd	Platinum Member
104	Noah	Brown	4567890123	noah.brown@example.com	321 Cedar Lane	Silver Member
105	Ava	Davis	5678901234	ava.davis@example.com	654 Birch Way	Gold Member

insert all into Library_Book

```
values (201, 'A Winter's Promise', 'Fantasy', 'Available', '15.99')
INTO Library_Book VALUES (202, 'Absolu', 'Romance', 'Available', '12.50')
INTO Library_Book VALUES (203, 'Scythe', 'Dystopian', 'Unavailable', '18.00')
INTO Library_Book VALUES (204, 'Divergent', 'Science Fiction', 'Available', '14.25')
INTO Library_Book VALUES (205, 'Harry Potter', 'Fantasy', 'Available', '22.00')
```

select 1 from dual;

BOOK_ID	TITLE	GENRE	AVAILABILITY	PRICE
201	A Winter's Promise	Fantasy	Available	15.99
202	Absolu	Romance	Available	12.50
203	Scythe	Dystopian	Unavailable	18.00
204	Divergent	Science Fiction	Available	14.25
205	Harry Potter	Fantasy	Available	22.00

insert all into Library_Preference

```
values (301, 'Fantasy')
INTO Library_Preference VALUES (302, 'Romance')
INTO Library_Preference VALUES (303, 'Dystopian')
INTO Library_Preference VALUES (304, 'Science Fiction')
INTO Library_Preference VALUES (305, 'Adventure')
```

select 1 from dual;

PREFERENCE_ID	PREFERENCE_TYPE
301	Fantasy
302	Romance
303	Dystopian
304	Science Fiction
305	Adventure

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```
insert all into Library_Librarian
values (401, 'Sophia Clark', 101)
INTO Library_Librarian VALUES (402, 'Jackson Miller', 102)
INTO Library_Librarian VALUES (403, 'Isabella Wilson', 103)
INTO Library_Librarian VALUES (404, 'Lucas Moore', 104)
INTO Library_Librarian VALUES (405, 'Mia Taylor', 105)
select 1 from dual;
```

LIBRARIAN_ID	NAME	MEMBER_ID
401	Sophia Clark	101
402	Jackson Miller	102
403	Isabella Wilson	103
404	Lucas Moore	104
405	Mia Taylor	105

```
insert all into Library_Book_Order
values (501, '1-Jan-2024', 'Yes', 'Pending', 101, 201)
INTO Library_Book_Order VALUES (502, '1-Jan-2024', 'No', 'Pending', 102, 202)
INTO Library_Book_Order VALUES (503, '1-Jan-2024', 'Yes', 'Delivered', 103, 203)
INTO Library_Book_Order VALUES (504, '1-Jan-2024', 'Yes', 'Pending', 104, 204)
INTO Library_Book_Order VALUES (505, '1-Jan-2024', 'No', 'Delivered', 105, 205)
select 1 from dual;
```

BOOK_ORDER_ID	ORDER_DATE	GUARANTEE_PAYMENT	STATUS	MEMBER_ID	BOOK_ID
501	1-Jan-2024	Yes	Pending	101	201
502	1-Jan-2024	No	Pending	102	202
503	1-Jan-2024	Yes	Delivered	103	203
504	1-Jan-2024	Yes	Pending	104	204
505	1-Jan-2024	No	Delivered	105	205

```
insert all into Library_Purchase
values (601, '1-Jan-2024', '15.99', 101, 201)
INTO Library_Purchase VALUES (602, '1-Jan-2024', '12.50', 102, 202)
INTO Library_Purchase VALUES (603, '1-Jan-2024', '18.00', 103, 203)
INTO Library_Purchase VALUES (604, '1-Jan-2024', '14.25', 104, 204)
INTO Library_Purchase VALUES (605, '1-Jan-2024', '22.00', 105, 205)
select 1 from dual;
```

PURCHASE_ID	PURCHASE_DATE	PAYMENT_AMOUNT	MEMBER_ID	BOOK_ID
601	1-Jan-2024	15.99	101	201
602	1-Jan-2024	12.50	102	202
603	1-Jan-2024	18.00	103	203
604	1-Jan-2024	14.25	104	204
605	1-Jan-2024	22.00	105	205

```
insert all into Library_Borrowing
values (705, '1-Jan-2024', '1-Jan-2024', 'No', 105, 205)
INTO Library_Borrowing VALUES (701, '1-Jan-2024', '1-Jan-2024', 'No', 101, 201)
INTO Library_Borrowing VALUES (702, '1-Jan-2024', '1-Jan-2024', 'No', 102, 202)
INTO Library_Borrowing VALUES (703, '1-Jan-2024', '1-Jan-2024', 'No', 103, 203)
INTO Library_Borrowing VALUES (704, '1-Jan-2024', '1-Jan-2024', 'No', 104, 204)
select 1 from dual;
```

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BORROWING_ID	DUE_DATE	RETURN_DATE	OVERDUE_STATUS	MEMBER_ID	BOOK_ID
705	1-Jan-2024	1-Jan-2024	No	105	205
701	1-Jan-2024	1-Jan-2024	No	101	201
702	1-Jan-2024	1-Jan-2024	No	102	202
703	1-Jan-2024	1-Jan-2024	No	103	203
704	1-Jan-2024	1-Jan-2024	No	104	204

insert all into Library_Reservation

values (801, '1-Jan-2024', 'Pending', 101, 201)

INTO Library_Reservation VALUES (802, '1-Jan-2024', 'Confirmed', 102, 202)

INTO Library_Reservation VALUES (803, '1-Jan-2024', 'Pending', 103, 203)

INTO Library_Reservation VALUES (804, '1-Jan-2024', 'Confirmed', 104, 204)

INTO Library_Reservation VALUES (805, '1-Jan-2024', 'Pending', 105, 205)

select 1 from dual;

RESERVATION_ID	REQUEST_DATE	STATUS	MEMBER_ID	BOOK_ID
801	1-Jan-2024	Pending	101	201
802	1-Jan-2024	Confirmed	102	202
803	1-Jan-2024	Pending	103	203
804	1-Jan-2024	Confirmed	104	204
805	1-Jan-2024	Pending	105	205

insert all into Library_Author

values (901, 'Christelle Dabos')

INTO Library_Author VALUES (902, 'Margot Dessenne')

INTO Library_Author VALUES (903, 'Neal Shusterman')

INTO Library_Author VALUES (904, 'Veronica Roth')

INTO Library_Author VALUES (905, 'J.K. Rowling')

select 1 from dual;

AUTHOR_ID	NAME
901	Christelle Dabos
902	Margot Dessenne
903	Neal Shusterman
904	Veronica Roth
905	J.K. Rowling

insert all into Library_Book_Author

values (901, 201)

INTO Library_Book_Author VALUES (902, 202)

INTO Library_Book_Author VALUES (903, 203)

INTO Library_Book_Author VALUES (904, 204)

INTO Library_Book_Author VALUES (905, 205)

select 1 from dual;

AUTHOR_ID	BOOK_ID
901	201
902	202
903	203
904	204